

COMP08101 Programming for Cyber Security

Coursework (Term 1: 2024)

Component 2 – Confirming DNS Ports (Weighting 60%)

Due Date 24th December 2024

Overview

Component 2 requires you to write scripts which can be chained to resolve whether an IP address is running a DNS server. For the purposes of this exercise, you should only check port 53.

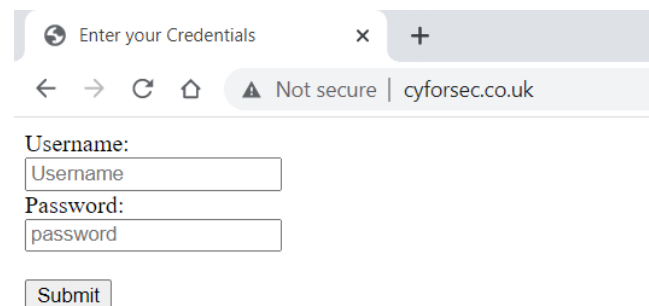
Learning Outcomes Addressed

- L1. effectively use a variety of software libraries to develop cyber security tools
- L2. employ software tools to facilitate and automate cyber security processes
- L3. Understand how software tools can be chained to accomplish complex tasks

Total Marks: 100 (Weightage: 60%)

Task 1 (25 marks)

You have been asked to gather information from the website <http://cyforsec.co.uk>.

A screenshot of a web browser window showing a login page for 'cyforsec.co.uk'. The browser's address bar shows the URL and a 'Not secure' warning. The page has a title bar that says 'Enter your Credentials'. There are two input fields: 'Username:' with the placeholder text 'Username' and 'Password:' with the placeholder text 'password'. Below these fields is a 'Submit' button.

The content on the site is protected by a form requiring a username and password. The username is '**admin**' and the password is a 4 digit code such as those commonly used to protect mobile phones. Valid examples of a passcode are 9012, 0313, 5168.

Using python, write a script **brute_force.py** to manipulate the URL and crack the passcode to reveal the contents of the file. You should adopt a brute force approach to this task, ie you should enumerate over (try) all the possibilities until you find the correct passcode. **You should not use libraries for this task.**

Submission

brute_force.py which should output the password for the login screen

Task 2 (25 Marks)

Once the password is cracked the contents of the webpage are revealed.

Login Succesful

```
13.66.139.0 - - [19/Dec/2020:13:57:26 +0100] "GET /index.php?option=
+http://www.bing.com/bingbot.htm" "-" 157.48.153.185 - - [19/Dec/2020
Gecko) Chrome/87.0.4280.88 Safari/537.36" "-" 157.48.153.185 - - [19/T
x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/87.0.4280.88 S
http://www.opensiteexplorer.org/dotbot, help@moz.com)" "-" 54.36.148.1
"Mozilla/5.0 (compatible; AhrefsBot/7.0; +http://ahrefs.com/robot/)" "-" S
SV1; .NET CLR 1.1.4322)" "-" 73.166.162.225 - - [19/Dec/2020:14:58:5
Chrome/87.0.4280.101 Safari/537.36" "-" 73.166.162.225 - - [19/Dec/202
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/87.0.4280.101 Safa
+http://ahrefs.com/robot/)" "-" 54.36.148.1 - - [19/Dec/2020:15:09:31 +0
AhrefsBot/7.0; +http://ahrefs.com/robot/)" "-" 162.158.203.24 - - [19/Dec
HTTP/1.1" 404 226 "-" "Mozilla/5.0 (compatible; Discorobot/2.0; +https:
8.1.0; zh-CN; EML-AL00 Build/HUAWEIEML-AL00) AppleWebKit/53
AliApp(DingTalk/4.5.11) com.alibaba.android.rimet/10487439 Channel/2
1725 "http://www.almhuetter-raith.at/" "Mozilla/5.0 (Linux; U; Android 8
baidu.sogo.uc.UCBrowser/11.9.4.974 UWS/2.13.1.48 Mobile Safari/537
/templates/_system/css/general.css HTTP/1.1" 404 239 "http://www.almh
Version/4.0 Chrome/57.0.2987.108 baidu.sogo.uc.UCBrowser/11.9.4.974
[19/Dec/2020:15:23:11 +0100] "GET /templates/jp_hotel/css/template.css
AppleWebKit/537.36 (KHTML, like Gecko) Version/4.0 Chrome/57.0.29
Channel/227200 language/zh-CN" "-" 42.236.10.117 - - [19/Dec/2020:15
EML-AL00 Build/HUAWEIEML-AL00) AppleWebKit/537.36 (KHTML
com.alibaba.android.rimet/10487439 Channel/227200 language/zh-CN" "
(Linux; U; Android 8.1.0; zh-CN; EML-AL00 Build/HUAWEIEML-AL0
AliApp(DingTalk/4.5.11) com.alibaba.android.rimet/10487439 Channel/2
```

Use the browser source code view to familiarise yourself with the structure of the page

```
Line wrap ☐
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title id="main">Login Succesful</title>
5 </head>
6 <body>
7 <h1>Login Succesful</h1>
8 <p class="logs">
9 13.66.139.0 - - [19/Dec/2020:13:57:26 +0100] "GET /index.php?option=cc
10 157.48.153.185 - - [19/Dec/2020:14:08:06 +0100] "GET /apache-log/acces
11 157.48.153.185 - - [19/Dec/2020:14:08:08 +0100] "GET /favicon.ico HTTP
12 216.244.66.230 - - [19/Dec/2020:14:14:26 +0100] "GET /robots.txt HTTP/
13 54.36.148.92 - - [19/Dec/2020:14:16:44 +0100] "GET /index.php?option=c
14 92.101.35.224 - - [19/Dec/2020:14:29:21 +0100] "GET /administrator/inc
15 73.166.162.225 - - [19/Dec/2020:14:58:59 +0100] "GET /apache-log/acces
16 73.166.162.225 - - [19/Dec/2020:14:58:59 +0100] "GET /favicon.ico HTTP
17 54.36.148.108 - - [19/Dec/2020:15:09:30 +0100] "GET /robots.txt HTTP/1
18 54.36.148.1 - - [19/Dec/2020:15:09:31 +0100] "GET /index.php?option=cc
19 162.158.203.24 - - [19/Dec/2020:15:16:50 +0100] "GET /apache-log/acces
20 35.237.4.214 - - [19/Dec/2020:15:22:40 +0100] "GET /administrator/%22
21 42.236.10.125 - - [19/Dec/2020:15:23:10 +0100] "GET / HTTP/1.1" 200 10
22 42.236.10.125 - - [19/Dec/2020:15:23:11 +0100] "GET /modules/mod_bowl
23 42.236.10.125 - - [19/Dec/2020:15:23:11 +0100] "GET /templates/_system
24 42.236.10.125 - - [19/Dec/2020:15:23:11 +0100] "GET /templates/jp_hote
25 42.236.10.117 - - [19/Dec/2020:15:23:11 +0100] "GET /templates/jp_hote
26 42.236.10.114 - - [19/Dec/2020:15:23:11 +0100] "GET /templates/jp_hote
27 42.236.10.114 - - [19/Dec/2020:15:23:11 +0100] "GET /templates/jp_hote
```

Using Beautiful Soup and Regular Expressions, write a script to scrape all the data contained in the `<p class="logs">` tag from cyforsec.co.uk. You should write the scraped content to a text file called **logs.txt**. You may wish to extend your **brute_force.py** script to perform this task or alternatively you can create a separate script.

Submission

- logs.txt, a text file containing the content scraped from cyforsec.co.uk
- brute_force.py or alternative. A python file used to scrape the data and write it to file.

Task 3 (25 Marks)

Using **logs.txt**, (once you have successfully written the data to the file you shouldn't need to keep hitting the server), write a script **extract_ips.py** that uses BeautifulSoup and Regular Expressions to extract all the valid IP addresses from **logs.txt**. Write the IP addresses to a text file **ips.txt** one per line.

Submission

- **extract_ips.py**, a Python script to extract valid IP addresses from **logs.txt**
- **ips.txt**, a text file containing the valid IP addresses.

Task 4 (25 Marks)

Using the `SynScan(host)` function introduced in the labs as a basis for your code write a script **confirm_dns.py** that reads each IP address in **ips.txt** and checks whether Port 53 is open

For each IP address you find with an open port 53 perform a reverse DNS check using the `ReverseDNS(ip)` function you were introduced to in the labs.

For IP addresses that are successfully resolved to a domain name. Use the `DNSScan()` function to check whether there is a DNS server running on Port 53. Write the results of your check to a text file **confirmed_dns.txt**. The output should look something like the following.

```
DNS Server at 54.36.148.92 : hydrogen092-ext2.a.ahrefs.com
DNS Server at 54.36.148.108 : hydrogen108-ext2.a.ahrefs.com
DNS Server at 54.36.148.1 : hydrogen001-ext2.a.ahrefs.com
DNS Server at 54.36.149.55 : hydrogen311-ext2.a.ahrefs.com
```

Submission

- **confirmed_dns.txt**, a text file containing the IP address and domain of IP addresses running a DNS server on Port 53
- **confirm_dns.py**, the Python script used to generate the results

Notes

Zip the files into a single compressed archive `<your_student_number>.zip` and submit through the link on Aula. Do make sure you answer the question that has been set, eg if you have been asked to write a function then make sure you write a function! Do comment your work and use appropriate naming conventions for any variables you declare. This assignment is an individual assessment and the work you submit should be your own.

If you are unable to break the password for the website, you may request the logs.txt file by emailing your Lecturer. This will enable you to attempt the other tasks. You will however receive limited or no marks for Task 1. You should email before the term ends on Friday 20th December. Any requests received after that will not be responded to.

Marking Scheme

Grade	Python file
A1 No improvement possible	
A2	All solutions are completed to outstanding standards reflecting a deep understanding of

	the programming techniques required. The code produced is professionally structured and makes full use of the programming techniques introduced on the module
A3	Fully completed to a professional standard; a clear understanding of the programming techniques required is obvious; The code produced is well structured and makes appropriate use of the techniques introduced on the module
B1	Completed to a high standard; an understanding of the programming techniques required is obvious. The code produced is structured and makes appropriate use of the techniques introduced on the module
B2	Solutions are created and largely correct; Code is structured but may have omissions or lack clarity
C	Solutions are created and mainly correct; Code is structured but may have omissions or lack clarity.
D	Solutions are created and all elements run although some output may be incorrect; Code is structured but may have omissions or lack clarity.
E	Solutions do not run or the output is largely incorrect. Code produced has no structure
N No attempt	